



Research articles

Approach to control magnetic plateau in tetrameric magnetic systems

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ABSTRACT

We study the magnetic behavior of a tetrameric magnetic system with four antiferromagnetic couplings (J_1 - J_2 - J_3 - J_4), where J_4 is the inter-cell coupling and J_1, J_2, J_3 are the intra-cell couplings, by using Green's function theory. The Magnetic-field curves shows that there are three plateaus ($M = 0, 1$ and saturated magnetization) and two gapless phases distinguished by four critical lines, and the plateaus widen or narrow with the inter-cell coupling (J_4), which are further explained using the energy spectrum. In addition, it is found that the magnetic contribution from each site of the unit cell is different due to the competition of the nearest neighboring pair interactions, which is demonstrated by the concurrence of the biparticle quantum state.

1. Introduction

Low-dimensional quantum spin systems have received considerable attention in condensed matter physics [1–12]. In particular, one-dimensional Heisenberg spin chains with competing interactions show a variety of physical properties that have been widely studied both theoretically [7–8] and experimentally [9–11,13].

In recent years, the Heisenberg spin chains with period $n = 2, 3$ and 4 have gained much attention, because they are the simplest quantum spin systems that can be used to simulate various quantum behaviors related to magnetization process, susceptibility and specific heat of materials [6–9,14–16]. In the presence of a magnetic field, magnetization plateaus appear in the Heisenberg spin chains when the energy gap is opened, which agree with the Haldane conjecture [17]. For the $S = 1/2$ compound $(\text{CH}_3)_2\text{CHNH}_3\text{CuCl}_3$ (IPACuCl₃) forming ferromagnetic (F)-antiferromagnetic (AF) alternating chain, and IPACuBr₃ forming AF-AF alternating chain, Manaka and Yamada have found that there is an energy gap between the singlet state and the excited state [9]. For spin-1/2 AF-AF-F and F-F-AF trimerized quantum chains, Gu and Su et al. numerically studied their magnetic behaviors and the specific heat by means of the transfer matrix renormalization group (TMRG) technique and the modified spin-wave (MSW) theory [6]. A magnetization plateau at $m = 1/6$ was observed in the two trimerized chains. In addition, the tetrameric Heisenberg spin chains with rich

quantum phenomena were also investigated, which includes F-AF₁-F-AF₂ [14], AF-AF-F [16], and AF₁-AF₂-AF₁-F chains [8,18]. The F-AF₁-F-AF₂ tetrameric chain can be obtained by a space modulation on the AF couplings of the AF-F Heisenberg spin model. Mahdavi et al. found that the space modulation induced a new gap in contrast to AF-F chains and the system experienced different quantum phases which were separated by four critical fields. Subsequently, another spin $S = 1/2$ tetrameric Heisenberg antiferromagnetic chain (HAFC) consisting of AF₁-AF₂-AF₃-AF₂ interactions were synthesized and studied theoretically [15]. It was found that there were five phase under the variation of the magnetic field. And the critical magnetic fields depend on exchange couplings. Because modulations of the magnetic interactions can easily be realized by using chemical modification easily [19–20], it is interesting to investigate the quantum behavior for tetrameric chains with different magnetic couplings (e.g. AF₁-AF₂-AF₃-AF₄).

2. Model and Hamiltonian

Yamaguchi et al first synthesized a single crystal of the verdazyl radical β -3-(2,6-dichlorophenyl)-1,5-diphenylverdazyl ($\beta - 2, 6 - \text{Cl}_2 - \text{V}$), and demonstrated it to be HAFC with fourfold periodicity by using ab initio molecular orbital calculation. The magnetic interactions can be modified by chemical modifications. It is of

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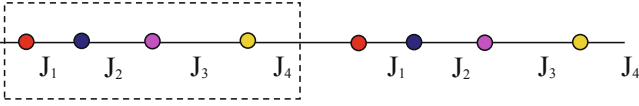


Fig. 1. The antiferromagnetic chain with fourfold periodicity. The interaction is J_1 - J_2 - J_3 - J_4 .

interest to investigate the different quantum effect for different J_4 . The sketch of the Heisenberg antiferromagnetic chain with fourfold periodicity are shown in Fig. 1. There are four interactions J_1 , J_2 , J_3 and J_4 in one magnetic unit cell. The Hamiltonian is as follows:

$$H = \sum_i [(J_1 \vec{S}_{4i+1} \cdot \vec{S}_{4i+2} + J_2 \vec{S}_{4i+2} \cdot \vec{S}_{4i+3} + J_3 \vec{S}_{4i+3} \cdot \vec{S}_{4i+4} + J_4 \vec{S}_{4i+4} \cdot \vec{S}_{4i+5}) - h(S_{4i+1}^z + S_{4i+2}^z + S_{4i+3}^z + S_{4i+4}^z)]$$

where $h = g\mu_B B$ is the reduced magnetic field giving rise to the Zeeman energy. J_i greater than 0 ($i = 1, 2, 3, 4$) are the HAFC nearest exchange couplings along the chain.

After the Jordan-Wigner (JW) transformation [21], the Hamiltonian (1) becomes

$$H = \frac{1}{N} \sum_i [\frac{1}{2}(J_1 a_i^\dagger b_i + J_2 b_i^\dagger c_i + J_3 c_i^\dagger d_i + J_4 d_i^\dagger a_{i+1} + h. c.) + J_1(a_i^\dagger a_i - \frac{1}{2})(b_i^\dagger b_i - \frac{1}{2}) + J_2(b_i^\dagger b_i - \frac{1}{2})(c_i^\dagger c_i - \frac{1}{2}) + J_3(c_i^\dagger c_i - \frac{1}{2})(d_i^\dagger d_i - \frac{1}{2}) + J_4(d_i^\dagger d_i - \frac{1}{2})(a_{i+1}^\dagger a_{i+1} - \frac{1}{2}) - h(a_i^\dagger a_i + b_i^\dagger b_i + c_i^\dagger c_i + d_i^\dagger d_i - 2)]$$

The Green's function method is employed to decouple the Hamiltonian[22], which is defined as

$$G_{ij}(t - t') = \langle \langle \alpha_i(t); \alpha_j^\dagger(t') \rangle \rangle = -i\theta(t - t') \langle \alpha_i \alpha_j^\dagger + \alpha_j^\dagger \alpha_i \rangle$$

$$\alpha = a, b, c, d$$

where the subscripts i and j label lattice sites. After the time Fourier transformation, the Green's function is put into the equation of motion,

$$\omega \langle \langle \alpha_i; \alpha_j^\dagger \rangle \rangle = \langle [\alpha_i; \alpha_j^\dagger] \rangle + [\alpha_i, H]; \alpha_j^\dagger$$

By performing the equation of motion analogous to Eq.(4) for the high-order Green's function $\langle \langle [\alpha_i, H]; \alpha_j^\dagger \rangle \rangle$, a higher-order Green's function will appear on the right side, resulting in an infinite set of coupled equations. By using bond-mean-field theory (BMFT), the quartic Ising term in Hamiltonian are treated to decouple scheme [5,23,24],

$$\alpha_i^\dagger \alpha_i \alpha_{i+1}^\dagger \alpha_{i+1} \approx \langle \alpha_i \alpha_{i+1}^\dagger \rangle \alpha_i^\dagger \alpha_{i+1} + \langle \alpha_{i+1} \alpha_i^\dagger \rangle \alpha_{i+1}^\dagger \alpha_i + \langle \alpha_i \alpha_{i+1}^\dagger \rangle \langle \alpha_{i+1} \alpha_i^\dagger \rangle$$

For further Fourier transformation into k -space, the Green's function can be expressed as

$$G_{ij} = \frac{1}{N} \sum_k g(k) e^{ik(i-j)}$$

The summation of the wave vector k is along the chain direction. So, the momentum space Green's function $g(k, \omega)$ can be characterized as a function of wave vector k and the elementary excitation spectrum $\omega = \omega(k)$.

After performing Eq.(5) and Eq.(6), the Eq.(4) would be described as

$$(\omega I - \Gamma)G = \langle [\alpha, \beta]_+ \rangle$$

where α, β correspond to a, b, c, d and $a^\dagger, b^\dagger, c^\dagger, d^\dagger$, respectively. And the left of Eq.(7) can be expressed as

$$\omega I - \Gamma = \begin{pmatrix} A_1 & J_1(q_1 - \frac{1}{2}) & 0 & J_2(q_4 - \frac{1}{2})e^{-ik} \\ J_1(q_1 - \frac{1}{2}) & A_2 & J_2(q_2 - \frac{1}{2}) & 0 \\ 0 & J_2(q_2 - \frac{1}{2}) & A_3 & J_3(q_3 - \frac{1}{2}) \\ J_2(q_4 - \frac{1}{2})e^{ik} & 0 & J_3(q_3 - \frac{1}{2}) & A_4 \end{pmatrix}$$

herein, A_1, A_2, A_3, A_4 are $\omega + h - J_4 m_d^z - J_1 m_b^z, \omega + h - J_2 m_c^z - J_1 m_a^z, \omega + h - J_2 m_b^z - J_3 m_d^z, \omega + h - J_4 m_a^z - J_3 m_c^z$, respectively and $m_a^z = \langle \alpha^\dagger \alpha \rangle - \frac{1}{2}$. And $q_1 = \langle b_n^\dagger a_n \rangle, q_2 = \langle c_n^\dagger b_n \rangle, q_3 = \langle d_n^\dagger a_n \rangle, q_4 = \langle a_{n+1}^\dagger d_n \rangle$, respectively.

According to the standard spectral theorem, the correlation function of the fermion operators can be calculated through the corresponding Green's function

$$\langle \alpha_j^\dagger \alpha_i \rangle = \frac{i}{2\pi N} \sum_k e^{ik(i-j)} \int \frac{d\omega}{e^{\beta\omega} + 1} [g(k, \omega + i0^+) - g(k, \omega - 0^+)]$$

where $\beta = \frac{1}{k_B T}$, k_B and T are the Boltzman's constant and the absolute temperature, respectively.

The average magnetization M per cell can be expressed as

$$M = \frac{1}{N} \sum_i (\langle m_{4i+1}^z \rangle + \langle m_{4i+2}^z \rangle + \langle m_{4i+3}^z \rangle + \langle m_{4i+4}^z \rangle)$$

We would like to mention that the phase transitions are signaled by the concurrence of biparticle quantum state[25,26].

$$C = \max\{0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\}$$

where $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are square roots of the eigenvalues of $\tilde{\rho}_{ij} \rho_{ij}$ in descending order. $\tilde{\rho}_{ij} = (\sigma_i^y \otimes \sigma_j^y) \rho_{ij}^* (\sigma_i^y \otimes \sigma_j^y)$. ρ_{ij}^* , σ^y is the complex conjugation of reduced density matrix ρ_{ij} and the y component of the Pauli operator. The $C = 0$ and $C = 1$ correspond to unentangled states and maximally entangled states, respectively. Herein, we define the concurrence C_{12} , C_{23} , and C_{34} connected by intra-cell interactions J_1 , J_2 , and J_3 , respectively. The inter-cell concurrence C_{41} is connected by J_4 .

Thereby, the above equations can be solved self-consistently with an initial state. In calculation, an initial state, composed of correlation functions between adjacent sites (q_1, q_2, q_3, q_4), and the average magnetizations at each site in one unit cell ($m_a^z, m_b^z, m_c^z, m_d^z$) are put into equations to produce resultant values. The iteration goes on until convergence is reached.

3. Results and discussion

In the following discussion, J_2 is taken to be an energy unit, and other parameters are $J_1 = 0.8 J_2, J_3 = 2 J_2$, the same as those in ref. [12]. In this paper, the results are obtained with the anisotropy parameter set to 0.05. First, we plot the magnetization curves under different magnetic fields for the situation where J_4 is less than 1.6 and J_4 is larger than 1.6 at zero temperature, respectively. To further understand the magnetic behavior, the spectral functions are shown. In addition, we calculate the contribution of each site in one unit cell to the total magnetization, and analyze the concurrence to characterize the critical field and the correlation between the two nearest sites.

In Fig. 2(a), the field dependences of the magnetization are plotted. Apparently, the appearance of three plateaus satisfies the Oshikawa, Yamanaka, and Affleck(OYA) theory [27]. There are five regions: three magnetic plateaus with 0, 1 and saturation magnetization plateau, and two gapless phases, similar to that of the tetrameric chain [12,15]. There are four critical magnetic fields (h_1, h_2, h_3 and h_4) distinguishing the five regions, which are characterized in Fig. 2. For J_4 equals 1.6, 0-magnetic plateau disappears. It can be explained by the spectral function with $J_4 = 1.6$ shown in the inset of Fig. 2(a). For the four energy branches, the two energy bands under the Fermi level are filled. The energy gap between energy branch ω_2 and ω_3 , which contributes to 0-magnetic plateau, becomes zero. Shown in Fig. 2(a), for J_4 is less than

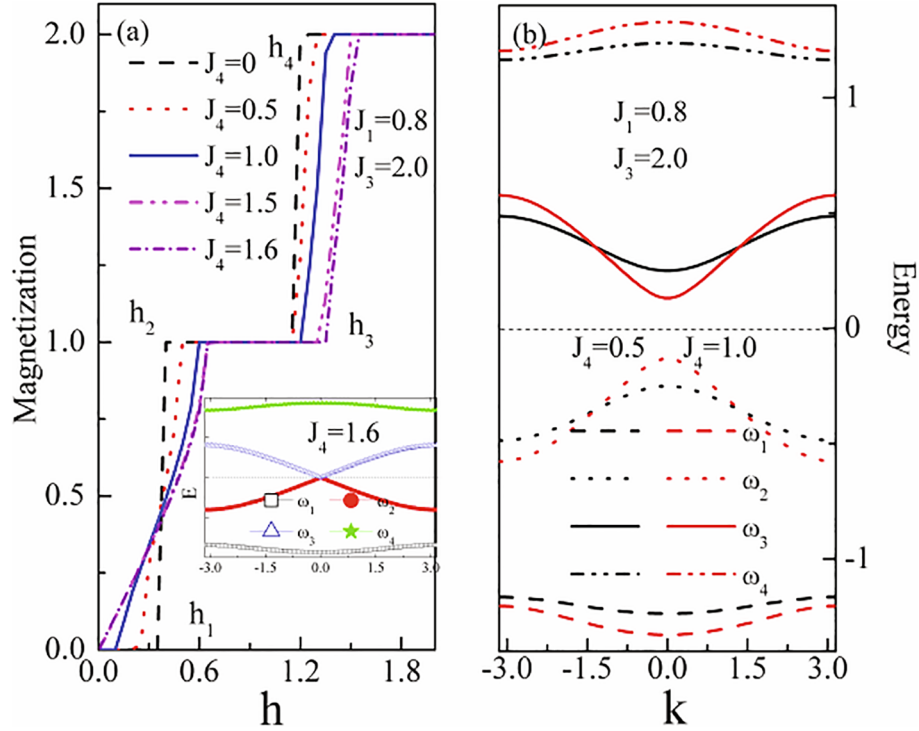


Fig. 2. (a) Magnetization curves under different magnetic fields for $J_4 = 0, 0.5, 1.0, 1.5$ and 1.6 , respectively; (b) The spectral function with $J_4 = 0.5$, and 1.0 . h_1, h_2, h_3 and h_4 characterize phase transition fields. The inset in Fig. 2(a): the spectral function for $J_4 = 1.6$. The dot line (energy equals zero) is Fermi level.

1.6, the second plateau does not change much. From this figure, it can also be seen that for $J_4 = 0.5$, the two gapless regimes (between h_1 and h_2 , and between h_3 and h_4) correspond to $0.2\text{--}0.5$ and $1.15\text{--}1.3$, respectively. While for $J_4 = 1.0$, the two gapless regimes correspond to $0.05\text{--}0.58$ and $1.2\text{--}1.4$. Through comparison, it can be obtained that the 0-magnetic plateau gets narrower and the two gapless regimes become wider for $J_4 = 1.0$ than that for $J_4 = 0.5$, which can be explained by the spectral function shown in Fig. 2(b). Comparing the energy

branches for $J_4 = 0.5$ with that for $J_4 = 1.0$, it is clearly seen that the energy branches get wider for $J_4 = 1.0$, resulting in the corresponding gapless phases getting wider. In addition, the gap between ω_2 and ω_3 become smaller, which are responsible for the 0-magnetic plateau.

As shown in Fig. 3(a), with J_4 increasing, 0-magnetic plateau widens, which can be obtained from the critical fields h_1 equal $0.03, 0.1, 0.15$ and 0.3 for J_4 set at $2.0, 2.4, 3.0$ and 5.0 , respectively. And the first gapless phase between h_1 and h_2 decreases. Especially, the second

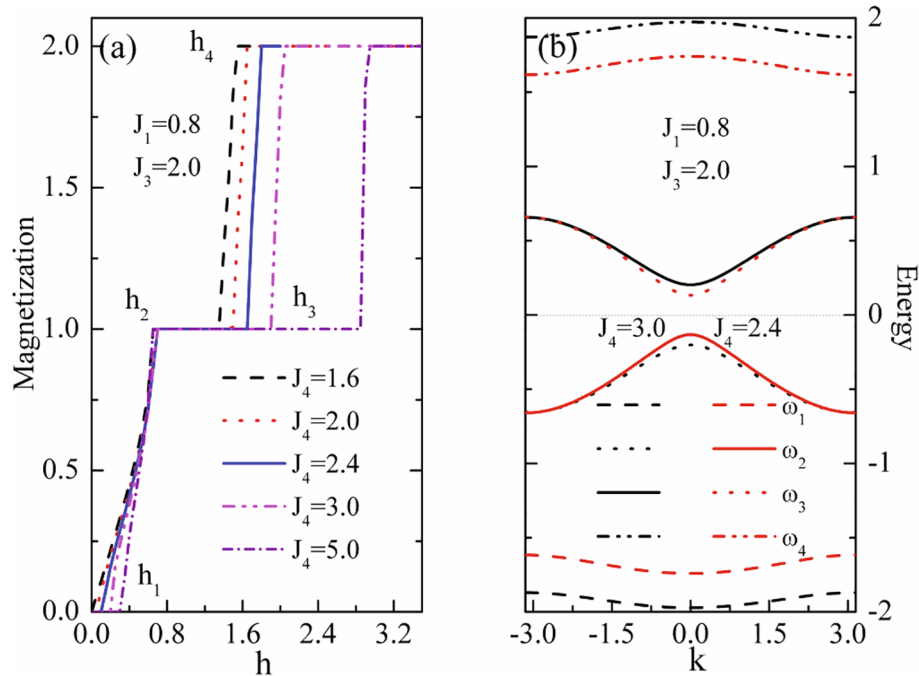


Fig. 3. (a) Magnetization curves under different magnetic fields for $J_4 = 1.6, 2.0, 2.4, 3.0$ and 5.0 , respectively; h_1, h_2, h_3 and h_4 characterize phase transition fields. (b) The spectral function with $J_4 = 2.4$, and 3.0 . The dot line (energy equals zero) is Fermi level.

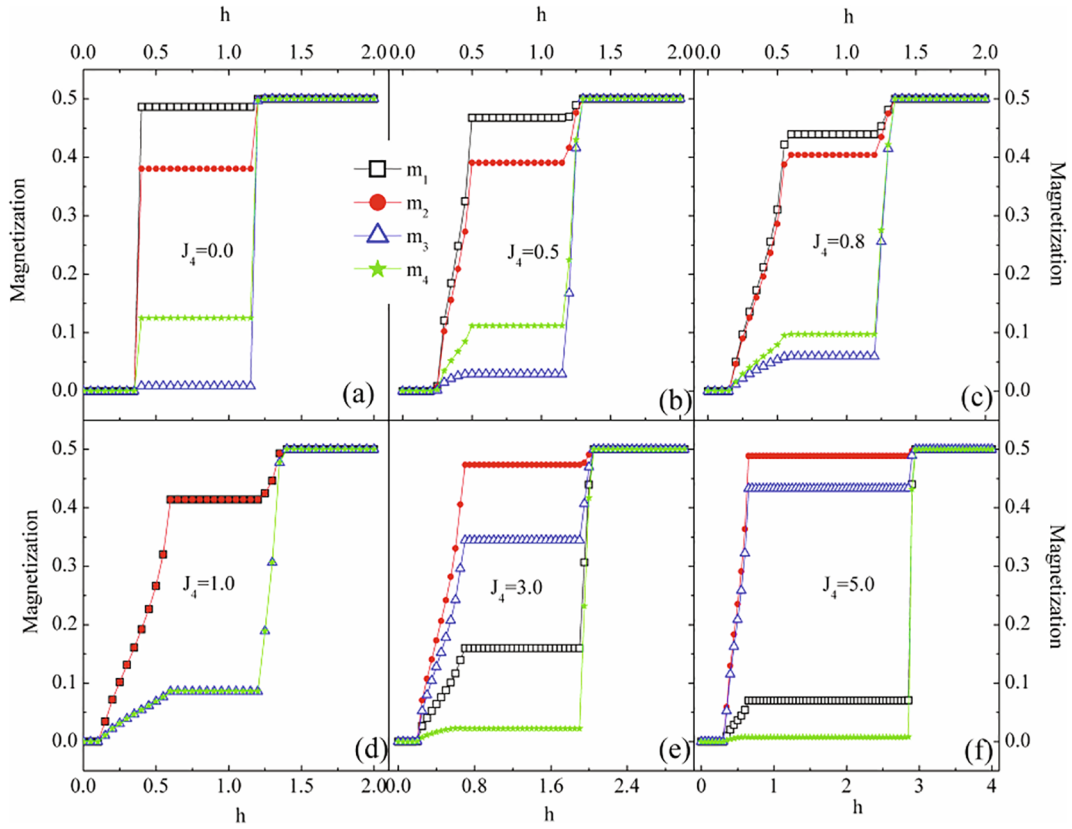


Fig. 4. The magnetization versus magnetic field h for (a) $J_4 = 0.0$; (b) $J_4 = 0.5$; (c) $J_4 = 0.8$; (d) $J_4 = 1.0$; (e) $J_4 = 3.0$; (f) $J_4 = 5.0$.

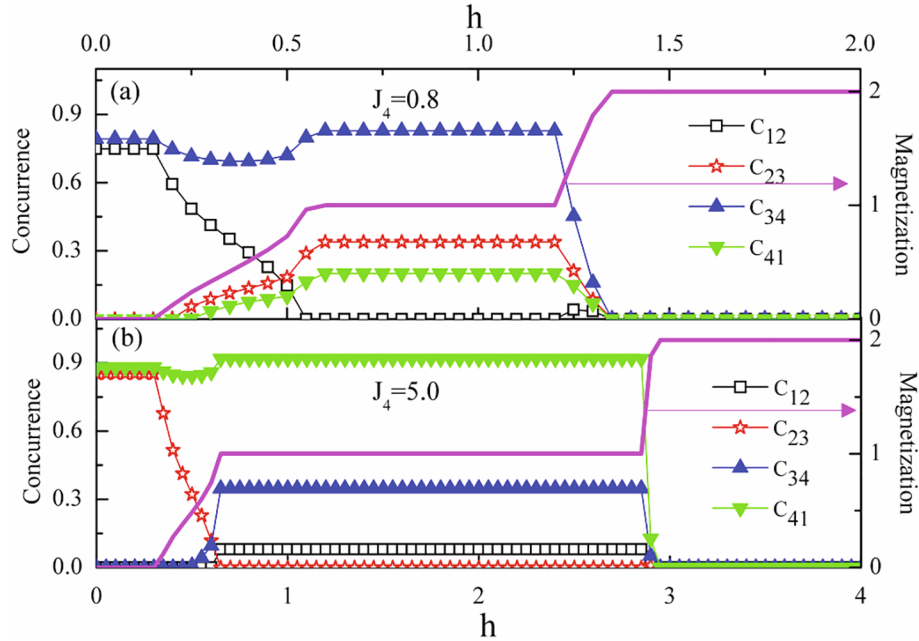


Fig. 5. The concurrence as a function of h for (a) $J_4 = 0.8$; (b) $J_4 = 5.0$.

plateau corresponding to the regime between h_2 and h_3 gets wider obviously as J_4 increasing. For $J_4 = 2.4$, the second plateau corresponds to the field regions 0.7–1.6, and for $J_4 = 3.0$, it corresponds to 0.7–1.9. It can be explained by the spectral function that there are tiny changes in the energy branches ω_3 and the obvious uplift of the energy branches ω_4 shown in Fig. 3(b).

For analyzing the field dependence of magnetization in detail, the magnetizations as a function of magnetic field h are shown in Fig. 4. For

J_4 less than 1.0, when the magnetic field corresponds to the second plateau among the five magnetic phases, the magnetization from m_1 and m_2 dominates over those from m_3 and m_4 , and m_1 and m_2 contribute to the second plateau predominately. For larger J_4 ($J_4 = 3.0$ and $J_4 = 5.0$), the magnetization from m_2 and m_3 dominates over those from m_1 and m_4 , and m_2 and m_3 contribute to the second plateau predominately. Fig. 5(a) and (b) plot the h dependence of the two-site concurrence. It is found that the concurrence dramatically changes at

the critical fields where the quantum phase transition happens. Moreover, for a strong field, all spins are parallel, and the spin for every site in a unit cell equals $1/2$.

To further understand the differences among the five magnetic regimes for different J_4 , the concurrence as a function of h is plotted in Fig. 5. For the magnetic fields where the magnetization plateaus appear, it can be found that the four concurrences display plateaus. As shown in Fig. 5(a), as the magnetic field is smaller than h_1 , the concurrences C_{34} and C_{12} are larger than C_{23} and C_{41} for $J_4 = 0.8$, which indicates that the magnetic field cannot destroy the singlet between the sites $4i + 3$ and $4i + 4$ and that between the sites $4i + 1$ and $4i + 2$. As the magnetic field increases, the singlet between the sites $4i + 1$ and $4i + 2$ are destroyed partially, while the singlet between the sites $4i + 3$ and $4i + 4$ is decomposed slightly. Until the magnetic field reaches h_2 , m_1 and m_2 prevail over m_3 and m_4 . Moreover, the neighbor interactions J_4 and J_2 also affect the magnetization of the sites $4i + 1$ and $4i + 2$, respectively, and as a result there is some difference between m_1 and m_2 . In the second magnetic plateau, the sites $4i + 3$ and $4i + 4$ almost reside in entangled state because the interaction (J_3) between the sites $4i + 3$ and $4i + 4$, is the largest interaction in the unit cell. As the magnetic field continue to increase, all the spins are parallel, and their concurrences exhibit plateaus.

In addition, for $J_4 = 5.0$, the concurrences C_{23} and C_{41} are larger than C_{34} and C_{12} , corresponding to the singlet between the sites $4i + 2$ and $4i + 3$ and that between the sites $4i + 4$ and $4i + 5$ (the first site of the next unit cell: $4(i + 1) + 1$). Since the interaction between the sites $4i + 2$ and $4i + 3$, are the smaller interaction, as a result, with the magnetic field increasing, the singlet between the sites $4i + 2$ and $4i + 3$ partially collapses and mainly contributes to the second plateau. Moreover, the neighbor interactions J_1 and J_3 , which are relevant to the sites $4i + 2$ and $4i + 3$, respectively, also play a role on m_2 and m_3 , and as a result there is some difference between m_2 and m_3 .

4. Conclusion

Using Green's function theory, we investigate the magnetic plateaus and critical behavior of a fourfold magnetic periodic system. We explore the magnetic field-induced quantum phase transitions for different inter-cell interaction (J_4). It is shown that when J_4 is less than 1.6, the 0-magnetic plateau narrows and the two gapless regimes widen with inter-cell interaction increasing, which can be explained by the widening of the energy branches near Fermi level. When $J_4 = 1.6$, 0-magnetic plateau disappears due to the closed energy gap near Fermi level. When further increasing J_4 , 0-magnetic and the second plateau widen, and the first gapless phase get smaller. In addition, in order to further understand the contribution of each site to the second plateau, we have calculated the concurrence of biparticle quantum state.

CRediT authorship contribution statement

Shuling Wang: Conceptualization, Methodology, Software, Data curation, Writing - original draft. **Denghui Ji:** Investigation, Formal

analysis. **Si-Cong Zhu:** Validation, Investigation. **Wei Zhao:** Investigation. **Kai-Lun Yao:** Conceptualization, Investigation. **Chi-Hang Lam:** Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jmmm.2020.167050>.

References

- [1] L.J. Ding, Y. Zhong, Phys. Chem. Chem. Phys. 20 (2018) 20228–20234.
- [2] G.S. Uhrig, H.J. Schulz, Phys. Rev. B 54 (1996) R9624–R9627.
- [3] A.B. Harris, Phys. Rev. B 7 (1973) 3166–3187.
- [4] T. Barnes, J. Riera, D.A. Tennant, Phys. Rev. B 59 (1999) 11384–11397.
- [5] L.J. Ding, Y. Zhong, S.W. Fan, L.Y. Zhu, Phys. Chem. Chem. Phys. 18 (2016) 510–518.
- [6] B. Gu, G. Su, S. Gao, Phys. Rev. B 73 (2006) 134427.
- [7] S.-S. Gong, B. Gu, G. Su, Phys. Lett. A 372 (2008) 2322–2325.
- [8] S.-S. Gong, G. Su, Phys. Rev. B 78 (2008) 104416.
- [9] H. Manaka, I. Yamada, J. Magn. Magn. Mater. 177–181 (1998) 681–682.
- [10] J.C. Livermore, R.D. Willett, R.M. Gaura, C.P. Landee, Inorg. Chem. 21 (1982) 1403–1405.
- [11] A.W. Garrett, S.E. Nagler, D.A. Tennant, B.C. Sales, T. Barnes, Phys. Rev. Lett. 79 (1997) 745–748.
- [12] S. Wang, R. Li, L. Ding, H.-H. Fu, S.-C. Zhu, Y. Ni, Y. Meng, K. Yao, J. Magn. Magn. Mater. 371 (2014) 77–81.
- [13] A. Escuer, R. Vicente, M.S. El Fallah, M.A.S. Goher, F.A. Mautner, Inorg. Chem. 37 (1998) 4466–4469.
- [14] M. Saeed, A. Jahanfar, J. Phys. Condens. Matter 23 (2011) 246002.
- [15] H. Yamaguchi, T. Okubo, K. Iwase, T. Ono, Y. Kono, S. Kittaka, T. Sakakibara, A. Matsuo, K. Kindo, Y. Hosokoshi, Phys. Rev. B 88 (2013) 174410.
- [16] H.T. Lu, Y.H. Su, L.Q. Sun, J. Chang, C.S. Liu, H.G. Luo, T. Xiang, Phys. Rev. B 71 (2005) 144426.
- [17] F.D.M. Haldane, Phys. Rev. Lett. 50 (1983) 1153–1156.
- [18] S.-S. Gong, S. Gao, G. Su, Phys. Rev. B 80 (2009) 014413.
- [19] K. Iwase, H. Yamaguchi, T. Ono, Y. Hosokoshi, T. Shimokawa, Y. Kono, S. Kittaka, T. Sakakibara, A. Matsuo, K. Kindo, Phys. Rev. B 88 (2013) 184431.
- [20] J. Zhang, K.L. Yao, Z.L. Liu, G.Y. Gao, X.C. Wang, Y.L. Li, Solid State Commun. 145 (2008) 43–47.
- [21] M. Azzouz, Phys. Rev. B 48 (1993) 6136–6140.
- [22] P. Fröbrich, P.J. Kuntz, Phys. Rep. 432 (2006) 223–304.
- [23] N. Zimbovskaya, Phys. Rev. B 78 (2008) 035331.
- [24] B.W. Ramakko, M. Azzouz, Phys. Rev. B 76 (2007) 064419.
- [25] W.K. Wootters, Phys. Rev. Lett. 80 (1998) 2245–2248.
- [26] R.-X. Li, S.-L. Wang, K.-L. Yao, H.-H. Fu, Phys. Lett. A 377 (2013) 2422–2427.
- [27] M. Oshikawa, M. Yamanaka, I. Affleck, Phys. Rev. Lett. 78 (1997) 1984–1987.